

WEEK 5 PROJECT SUMMARY

1. What I planned to accomplish in Week 5

Clean the appointment data and run the full exploratory analysis, calculate the five KPIs I proposed in Week 4, produce the charts and the initial analysis report, start the **cross-track exchange with the Data Science track**.

2. What I actually completed

The cleaning and quality checks, all five KPI calculations with interpretations, and the charts and report (Document 1). I also identified my collaboration point and prepared the files I will share (see point 7). The dashboard did not get started — I moved it to Week 6 (see point 6).

3. Key findings

The no-show rate is 48.5% (2,423 of 5,000 appointments). Cancellations are only 5.3%, so the clinic's problem is no-shows, not cancellations.

Booking lead time is the strongest driver: the no-show rate climbs from about 28% (0–7 days) to about 60% (31–60 days).

Patients with previous no-shows miss at 55.4%, against 43.5% for patients with a clean history.

Distance matters: the rate rises from 46.5% (0–10 km) to 68.1% (30+ km).

Reminders help less than I expected: only about 4 points (47.4% vs 51.4%), with SMS the best channel at 45.8%.

4. Major challenges

distance_to_clinic_km (90 blanks) and waiting_time_minutes (60 blanks) have missing values, and I had to decide not to impute them.

The weak reminder result contradicted my Week 4 assumption, so I re-ran the grouping twice before trusting it.

The dates arrived as M/D/YYYY although the dictionary says ISO, so I had to convert them before any time-based analysis.

5. Important decisions and why

Exclude cancellations from the no-show rate, because a cancellation frees the slot and a no-show wastes it.

Keep missing values as nulls instead of filling them, so I don't invent patterns.

Use bands (lead-time and distance) instead of raw values, because they are easier to read and act on.

Treat reminder_channel "None" as "no reminder sent," not as missing data.

Postpone the dashboard to Week 6 so this week's analysis stayed solid.

6. Changes to my Week 4 approach

Reminders moved from "main lever" to "modest effect"; lead time and patient history now lead my recommendations.

The dashboard moved from Week 5 to Week 6.

I added a reproducibility step (saving my cleaning steps as a script) after realising the Data Science track will need the same cleaned data.

7. Cross-track collaboration

Track: Data Science.

What I shared: the cleaned dataset with the lead-time and distance bands, my KPI definitions, and the no-show rates by band.

Their target definition, specifically whether they keep Cancelled separate or fold it into No-Show, so my KPIs and their model don't contradict each other.

Why it is relevant: we work on the same data; if our definitions differ, the project will produce two conflicting stories.

8. Proposed focus for Week 6

Build the interactive dashboard, complete and document the Data Science exchange, and refine the recommendations ahead of the testing and refinement stage.